# Project Initialization and Planning Phase

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| Date | 15 July 2026 |
| Team ID | SWTID-2026-2749 |
| Project Title | OptiCrop: Smart Agricultural Production Optimization engine |
| Maximum Marks | 5 Marks |

**Project Proposal (Proposed Solution) report**

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

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| **Project Overview** | |
| Objective | Develop an AI-powered crop recommendation system that predicts the most suitable crop using soil nutrients (N, P, K), temperature, humidity, pH, and rainfall. The objective is to improve crop productivity and support informed farming decisions. |
| Scope | The project provides crop recommendations for 22 crops using Machine Learning. It performs data analysis, preprocessing, model training, and real-time prediction through a Flask web application. |
| **Problem Statement** | |
| Description | Farmers often select crops based on traditional practices rather than scientific analysis, leading to poor productivity and financial loss. |
| Impact | Providing AI-based crop recommendations improves agricultural productivity, reduces farming risks, and supports sustainable agriculture. |
| **Proposed Solution** | |
| Approach | Develop a Flask-based web application integrated with a Random Forest Machine Learning model trained on crop recommendation data to predict the best crop. |
| Key Features | • AI-powered crop recommendation<br>• Random Forest prediction model |

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| | • User-friendly web interface<br>• Crop details (Season, Soil, Water Requirement)<br>• Responsive design |

**Resource Requirements**

| Resource Type | Description | Specification/Allocation |
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| **Hardware** | | |
| Computing Resources | CPU/GPU specifications, number of cores | Intel Core i5 or above |
| Memory | RAM specifications | 8 GB |
| Storage | Disk space for data, models, and logs | 20 GB Free Space |
| **Software** | | |
| Frameworks | Python frameworks | Flask |
| Libraries | Additional libraries | scikit-learn, pandas, numpy, matplotlib, seaborn |
| Development Environment | IDE | VS Code |
| **Data** | | |
| Data | Source, size, format | Crop Recommendation Dataset<br><br>2200 Records<br><br>CSV Format |